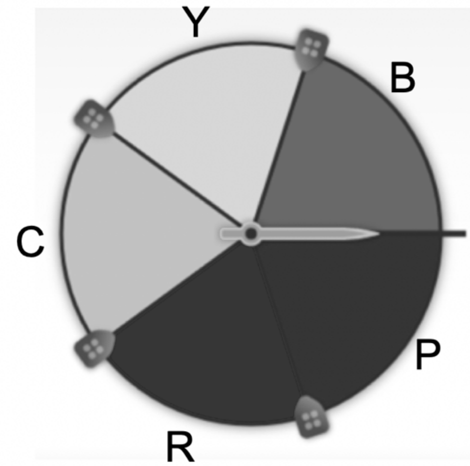
Grade 6 Accelerated
Unit 15
Lesson 6 Homework

Name_Answer Key________
Date____________________
Period__________________

1. Find the theoretical probability of landing on each color as a percentage rounded to the nearest tenth.

Color	Theoretical Probability
Cyan	$\frac{1}{5} = 20\%$
Yellow	20%
Blue	20%
Purple	20%
Red	20%



A computer simulator spinner performed more trials. The results as percentages are shown in the table below.

Color	Simulation 1 (10 trials)	Simulation 2 (100 trials)	Simulation 3 (1,000 trials)
Cyan	0%	19%	20%
Yellow	30%	23%	20%
Blue	30%	21%	20%
Purple	10%	16%	20%
Red	30%	21%	20%

2. What do you notice in comparing Simulation 1 to Simulation 2?

With more simulations, the probabilities for each color get closer to the theoretical probability.

3. Complete the table by estimating the percentages for Simulation 3.

Your teacher gives you a standard number cube with numbers 1–6 on its faces. You roll it 500 times. Your results are shown in the table below.

Outcome	1	2	3	4	5	6
Frequency	78	92	75	90	76	89

4. What is the experimental probability of rolling an even number?

$\frac{271}{500} = 0.542 = 54.2\%$

5. How many times would you expect to roll an even number?

$\frac{3}{6} = 0.5 = 50\%$ of the time

6. As the number of trials in this simple experiment increases, the
- o experimental probability
- o theoretical probability

gets closer to the
- o experimental probability.
- o theoretical probability.

Four situations are described. For each situation, create a possible simulation to fairly represent it.
Answers may vary.

7. Six people are going out to lunch together. One of them will be selected at random to choose which restaurant to go to. Who gets to choose?
Rolling a fair six-sided die to select which person will get to choose.

8. When a toddler walks, it is equally likely to step forward with its left foot or its right foot. Which foot will it use for its first step?
Flipping a coin where heads represent stepping forward with its left foot, and tail's represents stepping forward with its right foot.

9. Your school is taking 4 buses of students on a field trip. Will you be assigned to the same bus that your math teacher is riding on?
Spin a four colored spinner with one of the colors representing the bus your math teacher is riding on.

10. An experiment will produce one of ten different outcomes with equal probability for each. Why would using a standard number cube to simulate the experiment be a bad choice?
The experiment has ten possible outcomes, while a standard number cube has only six possible outcomes.